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RESEARCH ARTICLE

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Active wake control: An approach to optimize the lifetime operation of wind farms

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Abstract

The wind turbines within a wind farm impact each other's power production and loads through their wakes. Wake control strategies, aiming to reduce wake effects, receive increasing interest by both the research community and the industry. A number of recent simulation studies with high fidelity wake models indicate that wake mitigation control is a very promising concept for increasing the power production of a wind farm and/or reducing the fatigue loading on wind turbines' components. The purpose of this paper is to study the benefits of wake mitigation control in terms of lifetime power production and fatigue loading on several existing full-scale commercial wind farms with different scale, layouts, and turbine sizes. For modeling the wake interactions, Energy Research Centre of the Netherlands' FarmFlow software is used: a 3D parabolized Navier-Stokes code, including a k-ε turbulence model. In addition, an optimization approach is proposed that maximizes the lifetime power production, thereby incorporating the fatigue loads into the optimization criterion in terms of a lifetime extension factor.

KEYWORDS

active wake control, fatigue loads reduction, power production maximization, wake mitigation, wind farm control

1 | INTRODUCTION

Wind turbines are being densely clustered in wind farms and effect each other's power production and loading through their wakes. Because of the reduced wind velocity and increased turbulence, a turbine operating in the wake of another turbine will produce less power and experience increased fatigue loading. Still, the common practice is to let each wind turbine maximize its individual power capture, thereby disregarding its effect on other turbines. This "greedy" approach is not optimal with respect to the power production of the whole wind farm. At the beginning of this century, researchers started the development of a cooperative approach to operate wind farms[1-3] that aim at maximizing the power production of the whole farm, while at the same time, trying to reduce the fatigue loading on the wind turbines. These methods for mitigation of wake effects are called here *active wake control* (AWC).

There are 2 major classes of AWC methods. The first concept aims at reducing the wake deficit downstream by reducing the axial induction factor of upstream wind turbines, known as axial induction control (also referred to as heat and flux or pitch-based AWC). This is achieved by operating the turbines at the windward side at an increased blade pitch angle.[4,5] In practice, increased pitch angle could indirectly be achieved by reducing the power production of the turbine (derating). While the power production of the derated machines decreases, the wind velocity in their wakes increases, allowing to produce more power at downstream wind turbines, and, possibly, raises the overall farm production. Initial experiments in a wind tunnel[5,6] and in the field[7] have shown that the power production of a wind farm may potentially increase under pitch-based AWC. However, more-recent high-fidelity simulations[8] and wind tunnel tests[9] were unable to confirm these power benefits, which could be because of several reasons (see, eg, the discussions in Boorsma, [7] Campagnolo et al,[9] and Ciri et al[10]). Even though these results question the feasibility of power increase with pitch-based AWC and call for proper full-scale testing, the benefits with respect to load reduction are still considered promising. The other class of AWC methods is to redirect the wakes aside from downstream wind turbines. Wake redirection can be achieved by operating the wind

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turbines at yaw misalignment (yaw-based AWC, sometimes referred to as controlling wind),[11] although other approaches have also been studied in the literature: rotor tilting, yaw-oriented and tilt-oriented individual pitch control, and even wind turbine repositioning.[12-14]

Active wake control is attracting increasing attention in the literature during the past several years. Several publications are focused on the application of advanced optimization methods to solve the power production maximization problem and do not consider the loads. Examples are game-theoretic methods, which have been applied to both pitch-based[15] and yaw-based AWC,[16-18] nonlinear optimization methods, [19] extremum-seeking control,[20] and maximum power point tracking.[21,22] The game-theoretic approaches are capable to getting very close to the global optimum but require a huge number of iterations to converge, making them only suitable in combination with very simple wake models. The extremum-seeking control and maximum power point tracking approaches belong to the class of model-free methods and optimize the blade pitch and/or yaw misalignment angles directly based on the power measurements; however, they perform well only at very low turbulence levels and are difficult to implement in realistic environmental conditions.[20] Other publications aim at reduction of the fatigue loads on the turbines' components during periods of power curtailment, addressing the corresponding problem using optimization methods,[2,23-26] H_2 control,[27,28] and model predictive control.[29,30] These methods address the fatigue loading rather indirectly.

The majority of these publications are focused on the optimization methods while using very simplistic wake models, such as the Jenson[31] or the FLORIS model.[17,32] While the accuracy of these simple models can significantly be improved by including uncertainty in the wind direction,[33] some wake model parameters need to be fitted from data (from, eg, high-fidelity simulations) for a specific farm application. High-fidelity models based on computational fluid dynamics, such as SOWFA,[34] are typically only used to simulate very simplified setups consisting of just a few wind turbines for a limited set of wind velocities and directions.[12-14,17,18] While such tools are very useful for analysis, their computational complexity is prohibitive for even calculating the yearly power production of a medium-sized wind farm, let alone performing AWC optimizations for the lifetime power production. To achieve a suitable (for the AWC application) trade-off between model fidelity and computational load, Energy Research Centre of the Netherlands (ECN) has developed a 3D parabolized Navier-Stokes code FarmFlow, that uses a $k - \varepsilon$ turbulence model to account for turbulent processes in the wake. Atmospheric stability is accounted for via the Monin-Obukhov length parameter. The wake model is validated on measurement data from different full-scale wind farms[35] and delivers very accurate predictions of the power productions of the wind turbines at reasonable computational load: A calculation of a single wake takes about 2 seconds on a single core, and all available cores can be used automatically.

The purpose of this paper is to demonstrate the benefits of AWC in terms of the lifetime power production and lifetime fatigue loading for different real-life wind farms. The main contributions with respect to previous work are as follows: (1) The optimization uses a validated wake model (FarmFlow), (2) the effect of AWC on the complete lifetime is considered instead of just a few wind conditions, (3) the effect on the lifetime fatigue loading is included into the optimization and analysis, and (4) the evaluation is performed on several existing commercial wind farms. A numerically efficient optimization approach that uses the FarmFlow model to obtain the power productions of the turbines is used. In addition, a database is used to interpolate the fatigue loads on the main components of each turbine using the wind and wake conditions calculated by FarmFlow. To construct the loads database, simulations have been performed with a generic 5-MW wind turbine model using Focus/Phatas. Even though the underlying BEM theory is generally considered as inaccurate for load modeling under yawed conditions, recent studies[36] indicate that it captures the load trends sufficiently well. As the present application only requires the relative effects of AWC strategies on the fatigue loading, using standard BEM theory is considered sufficient. The cost function in the optimization is the lifetime power production, which includes both the yearly power production and the fatigue loading. Since the relationship between fatigue loads and operation and maintenance (O&M) costs is difficult to model, fatigue loads are considered through their effect on the lifetime of the turbine components. The results presented indicate that for AWC can potentially increase the power production of a wind farm by more than 1% while, at the same time, reducing the fatigue loading.

The paper is organized as follows. In the next section, the FarmFlow wake model used in this work is described. Section 3 describes the optimization approach in details. Several case studies with full-scale commercial wind farms are next presented in Section 4. The paper is concluded with some final remarks in Section 5.

2 | WAKE MODELING IN FARMFLOW

The AWC approach described in this paper uses ECN's FarmFlow software, an advanced and validated tool for the calculation of wake effects of wind farms. FarmFlow computes the average wind speed and the turbulence intensity in the wake of each wind turbine. The wake model in FarmFlow is a 3D parabolized Navier-Stokes code, using a k-ε turbulence model that accounts for turbulent processes in the far wake (k is defined as the turbulent kinetic energy and ε, its dissipation rate). The turbulence model describes the production, transport, and dissipation of k and ε. The ambient flow is modeled in accordance with the method of Panofsky and Dutton.[37] The free stream wind as a function of height is calculated for a prescribed ambient turbulence intensity and Monin-Obukhov length, which takes the atmospheric stability into account. In the model, the wake is divided in a near-wake region with a length of 2.25 rotor diameters (D) and a far-wake region. Because of the parabolization, axial pressure gradients are neglected. Since the flow deceleration and wake expansion in the near wake are forced by axial pressure gradients, the modeling in the far wake starts with an initial wake velocity profile at 2.25D.

The wake model has been improved by prescribing the streamwise pressure gradient as a source term in the flow equations. The streamwise pressure gradients are calculated via an inviscid, axisymmetric, free vortex wake method. With this method, the pressure gradients are a function of the axial force coefficient only. To save computational effort, the pressure gradients are calculated a priori for a large number of axial induction factors, and interpolation is used for other values. This hybrid method of wake modeling in the near-wake region, including an adapted near-wake turbulence model, gives very accurate results in an acceptable amount of computational time. Finally, the model allows to include wind direction variations and meandering effects by means of wind direction uncertainty distribution.

The code FarmFlow has been validated in Bot[35] using a large amount of experimental data from the ECN test site at Wieringermeer. Additionally, experimental data from 3 large offshore wind farms have been compared with FarmFlow model results. The calculated wake velocity deficits and turbulence intensities agree very well with experimental data for all wind speeds and ambient turbulence intensities. Excellent agreement between calculated and measured turbine performance is found for the large offshore wind farms Horns Rev and Nysted.

3 | AWC APPROACH

As already discussed in Section 1, there are 2 classes of AWC methods:

1. Pitch-based AWC: The axial induction of upstream turbines is reduced at below rated farm power (typically by increasing the blade pitch angle, or derating). This results in the velocity of the wake increasing, allowing downstream turbines to increase their power production. The turbulence intensity in the wake decreases as well, leading to lower fatigue loads downstream.
2. Yaw-based AWC: By operating upstream turbines at yaw misalignment, their wakes are redirected away from the downstream turbines, letting them produce more power. Because of the yaw misalignment, yaw-based AWC could potentially increase the loads for some wind directions. Therefore, it is important to be able to evaluate the effect of AWC on the fatigue loads and even include loads into the optimization to ensure they remain lower than the nominal loads. This will be discussed in more details in the sequel.

3.1 | Optimization criterion

The AWC farm operation strategy has influence on the power production of the farm and on the loads of the wind turbines. The most straightforward goal is to maximize the yearly power production of the wind farm

$$J_{1yr}(\theta^{AWC}, \phi^{AWC}) = \sum_{V} \sum_{\alpha} p(V, \alpha) P(V, \alpha, \theta^{AWC}, \phi^{AWC}), \tag{1}$$

wherein $p(V, \alpha)$ is the discrete probability density function, V is the undisturbed wind speed, α is the wind direction, and

$$P(V, \alpha, \theta^{AWC}, \phi^{AWC}) = \sum_{i=1}^{N} P_i(V, \alpha, \theta_i^{AWC}, \phi_i^{AWC})$$

with P_i being the power production of the ith turbine, including wake losses, for wind speed V, direction α, pitch angle bias (θ_i^{AWC}), and yaw misalignment (ϕ_i^{AWC}).

Considering the effect of the loads is more difficult because of the uncertain relationship between loads and O&M costs. Therefore, it is proposed here to include loads by means of their effect on the lifetime of components, assuming that it is possible to extend the lifetime of the wind farm. According to the widely used inverse power law, the lifetime L and stress S are related as $L(S) = \frac{1}{KS^M}$, K and M being parameters describing the material properties.[38] The loads are considered at design critical on the turbine, and the lowest loads reduction is used to determine the effect on the turbine lifetime.

Let, in the nominal case (without AWC), $S_{i,l}$ be the damage equivalent load at location l, $l = \overline{1, L}$, for the ith turbine, $i = \overline{1, N}$. If under AWC this loads becomes $S_{i,l}^{AWC} = S_{i,l}$, the lifetime at each location changes by a factor of $\left(\frac{S_{i,l}}{S_{i,l}^{AWC}}\right)^M$. Defining the lifetime extension factor for the whole farm as

$$f_{life}(\theta^{AWC}, \phi^{AWC}) = \min_{l \in 1}^{L} \left(\frac{1}{N} \sum_{i=1}^{N} \left(\frac{S_{i,l}}{S_{i,l}^{AWC}} \right)^M \right), \tag{2}$$

the optimization criterion including loads becomes the farm power production over the (extended) lifetime:

$$J_{life}(\theta^{AWC}, \phi^{AWC}) = f_{life}(\theta^{AWC}, \phi^{AWC}) J_{1yr}(\theta^{AWC}, \phi^{AWC}). \tag{3}$$

To evaluate the yearly production of the farm $J_{1yr}(\theta^{AWC}, \phi^{AWC})$, FarmFlow software is used. For the estimation of the lifetime in Equation 2, a precomputed-loads database is used. The database is constructed using aeroelastic simulations with a generic 5-MW wind turbine under many

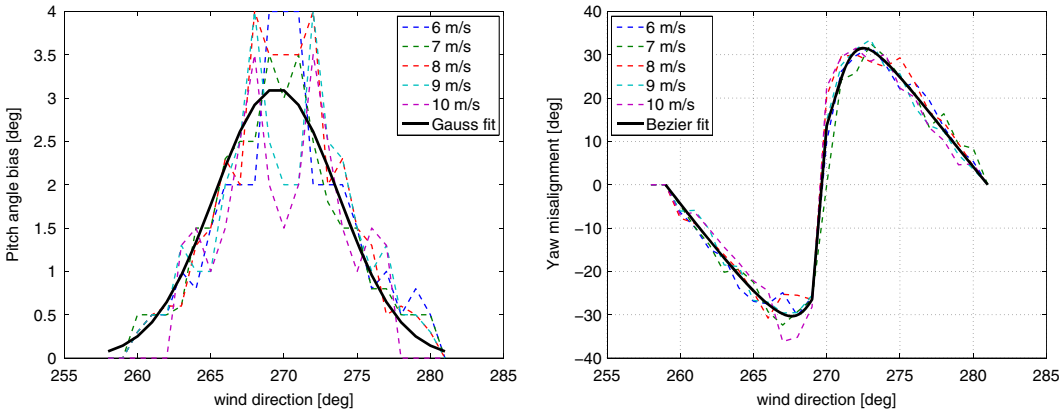


FIGURE 1 Curve fitting for typical active wake control settings over a sector of wind directions [Colour figure can be viewed at wileyonlinelibrary.com]

different operating conditions (wind speeds, turbulences, wake profiles, wake locations, and yaw misalignments). However, the database does not include the effect of pitch angle variations on the loads (yet). Therefore, pitch-based AWC is only applied to maximize the yearly power production in the case studies in this paper. Notice that derating by pitching generally reduces the fatigue loads.[39] The effect of AWC on the pitch and yaw actuator loads are subject to ongoing research; some preliminary results can be found in Kanev. [40]

In the following sections, it is explained how the 2 optimization problems (1) and (3) are addressed in a numerically efficient manner.

3.2 | Selected approach and assumptions

For each wind direction and wind speed, the optimal AWC settings (pitch angles and yaw misalignment angles) are determined for all turbines that have a turbine in their wake. Based on past experience, AWC is not applied for wind directions for which the distances between the wind turbines are too large (more than 20D), because AWC delivers practically no benefits in such cases. Since small changes in the wind direction can have large effects on the AWC settings, the wind directions are binned at 1 degree. The output from the optimization is a 3D table containing the pitch angle offsets or yaw misalignments for each wind turbine in the park for the considered wind speeds and directions.

3.2.1 | When loads are not included

For the yearly power production maximization (1), the AWC settings can be optimized for each wind direction independently (called here single direction–based optimization).

Furthermore, the pitch-based AWC optimization is restricted just to the so-called leading wind turbines, ie, wind turbines that are not in the wake of other turbines but have turbines in their wakes. The reason for that is that the turbulence intensity in the wake is too high, leading to faster wake recovery. The potential power increase behind the second turbine is relatively small compared with the power loss when pitch-based AWC is applied to it. This, however, does not hold for yaw-based AWC.

To further reduce the computational complexity, 2 simplifications are made:

- Row-by-row optimization: For a given wind direction, the rows of turbines that are in each other's wake are determined. For each row, a separate AWC optimization is performed.
- At most 2 variables per row: In each row, pitch-based AWC will be applied to just the first (leading) turbine. Yaw-based AWC, however, is applied to all turbines in the row but the second last one. Past experience shows that for a row of wind turbines yaw-based AWC optimization often results in an almost linear distribution of the yaw angles over the row. Hence, to reduce the number of optimization variables, the yaw angles within a row are restricted to be linearly dependent on the turbine number in the row. This results in just 2 optimization variables per row: the yaw angles of the leading and the second last turbine in the row.

3.2.2 | When loads are included

The optimization criterion including loads (3) involves the lifetime extension factor f_{life}. Strictly speaking, this optimization will then need to be solved in 1 shot, considering all wind speeds and directions and the corresponding settings of all wind turbines. However, the complexity of such an optimization is computationally prohibitive. Instead, a *sector-based optimization* approach is proposed, ie, for a row of turbines, a sector of wind directions is identified for which the optimization over the AWC settings is performed.

Based on numerous simulation studies, it is concluded that the optimal AWC settings only weakly depend on the wind speeds but strongly depend on the wind directions. Therefore, to further reduce computational complexity, the AWC settings are constrained to have predefined form as function of the wind within inside a given direction sector. The forms of these curves have been selected by performing numerous unconstrained optimizations for the pitch and yaw angles on a row of turbines. This analysis motivates the following choices:

- Pitch-based AWC: The dependency of the pitch angle offset on the wind direction is approximated by a Gaussian curve centered at the direction of the row of turbines. This choice is motivated from Figure 1 (left), showing that the pitch angle offset for the first turbine in a row optimized for different wind speeds and directions (dashed curves) are well approximated by a Gaussian curve (solid line) that depends on the wind direction only. The seemingly large variations in the optimal pitch angle for different wind speeds are primarily due to convergences to local optima and have very small effect on the final power production benefit.
- Yaw-based AWC: A good approximation of the optimal yaw angles as functions of the wind directions is obtained using 2 quadratic Bézier curves, left and right with respect to the direction of the row of turbines. This approximation with 2 Bézier curves is, of course, valid for 1 single wind turbine. As yaw-based AWC will need to optimize the yaw misalignment settings for the leading and the second last wind turbine, there will be 2 such pairs of Bézier curves required for 1 sector-based yaw-based AWC optimization.

4 | CASE STUDIES

4.1 | Considered wind farms

To demonstrate the benefits from AWC with respect to lifetime power production and lifetime fatigue loads, results are presented from AWC optimizations for several full-scale commercial wind farms. These farms are summarized in Table 1, wherein for each farm, its location (onshore or offshore), the approximate wind farm power capacity, the wake losses, the average distance (DIST) between turbines expressed in number of rotor diameters D, and the power density (power capacity divided by the area of the farm) are indicated. Because of confidentiality, the names of most of the wind farms and their exact power capacities are not disclosed. Since some of the analyzed wind farms do not have a regular layout and the distances between the turbines have large variations, an average distance is calculated that results when the same number of turbines are placed in a square area of the same size as the farm area, ie,

$$\text{DIST} = \frac{\sqrt{\text{farm area}}}{\sqrt{\text{nr. turbines}} - 1}.$$

The wide variety of farms considered allows to get a good understanding of the effects of different farm parameters on the benefits of AWC.

In what follows, first, a detailed analysis is presented for the Princess Amalia Wind Farm. Subsequently, the overall results are summarized and discussed for all considered farms.

4.2 | Detailed analysis of the results for the Princess Amalia Wind Farm

This section zooms in on the Princess Amalia Wind Farm and presents a more detailed analysis and discussion. The wind farm is located in the North Sea at 23 km from the village of Egmond aan Zee, The Netherlands. For this farm, pitch-based and yaw-based AWC have been optimized separately. Combination of the 2 strategies remains a topic of future research.

4.2.1 | Princess Amalia layout and setup

The wind farm consists of 60 Vestas V80 (2 MW) wind turbines. However, in this study, a generic 5-MW wind turbine model is used, a slight modification of the UpWind 5-MW reference wind turbine. The wind farm layout is scaled so that the distances between the wind turbines remain unchanged in terms of rotor diameters.

TABLE 1 Wind farms information

Wind farm	Location	Appr. Capacity, MW	Wake loss, %	DIST, D	Power density, MW/km^2
Wind farm 1	Offshore	300	9.57	7.93	8.41
Wind farm 2	Offshore	400	10.40	6.54	7.67
Wind farm 3	Onshore	30	10.39	4.96	17.34
Princess Amalia	Offshore	120	11.13	6.99	8.44
Wind farm 5	Onshore	200	9.95	7.35	5.08
Wind farm 6	Offshore	300	12.64	6.33	9.84

The wind speed, direction, and turbulence intensity distributions for this site are calculated from measurement data of the nearby Offshore Wind farm Egmond ann Zee (OWEZ) meteorological mast. The Weibull distribution parameters that give the best least squares fit to the measurements are $A = 7.63$ and $k = 2.53$. The farm layout is given in Figure 2.

4.2.2 | Pitch-based AWC

To get a better understanding of the benefits of pitch-based AWC, let us first inspect the results for 1 single row of wind turbines, namely, turbines T56 to T3. For this row, consider first the situation that the wind comes from a direction of 179° (ie, from the South to North, along the row of wind turbines T56 to T3) with a velocity of 8 m/s. These conditions are optimal with respect to the achievable benefit from pitch-based AWC for this row of turbines. For these conditions, Figure 3 gives the power productions and the turbulence intensity per wind turbine, both in the reference case (blue bars) and with pitch-based AWC (red bars). The corresponding pitch angle settings are given in the left plot of Figure 4 (notice that the nominal pitch angle is 0.4°). As expected, the leading turbine produces less power because of the increased pitch angle, while the remaining wind turbines increase their output. As a result, the total power production increases with for row of wind turbines T56 to T3 with 2.2% for the considered wind direction (179°) and wind speed (8 m/s). Looking at the whole park, the power increase reduces to 2.0% (still for 1 single wind direction and wind speed). Considering that pitch-based AWC has effect only at below rated farm power, using the wind speed distribution, the yearly increase of the power production for the whole wind farm for this particular wind direction drops to 0.88%. The relative yearly power production increase is summarized in the left plot of Figure 5 as function of the wind direction.

Weighing the relative power increase per wind direction with the wind direction probabilities, the total yearly power increase for the whole farm becomes around 0.33%. Expressed in terms of wake loss reduction, around 3% of the wake losses is regained by pitch-based AWC. The right-hand–side plot in Figure 5 exemplifies the pitch angle offset as the function of the wind direction for 2 turbines located at the edge of the farm,

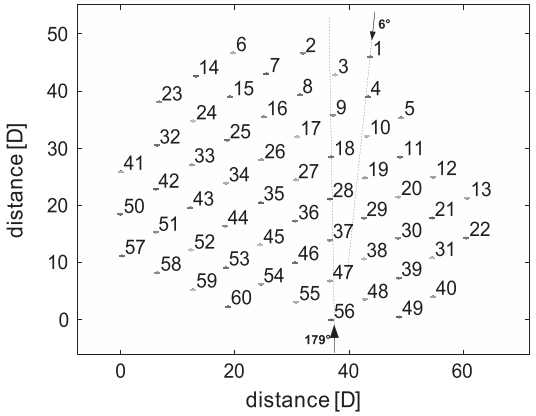


FIGURE 2 Layout of the Princess Amalia offshore wind farm

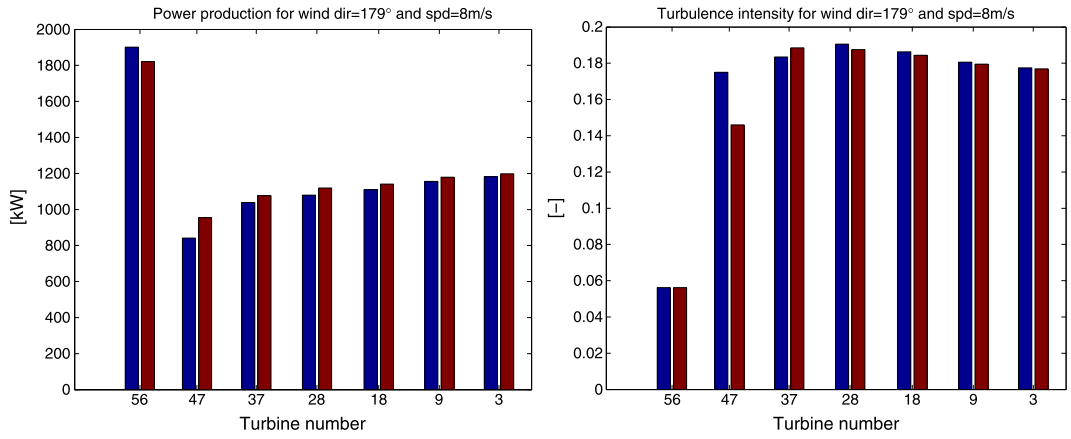


FIGURE 3 Power production (left) and turbulence intensity (right) for the row of wind turbines T56 to T3 for wind direction 179° and wind speed 8 m/s. The blue (left) bars represent the reference case, while the red (right) bars represent pitch-based AWC results [Colour figure can be viewed at wileyonlinelibrary.com]

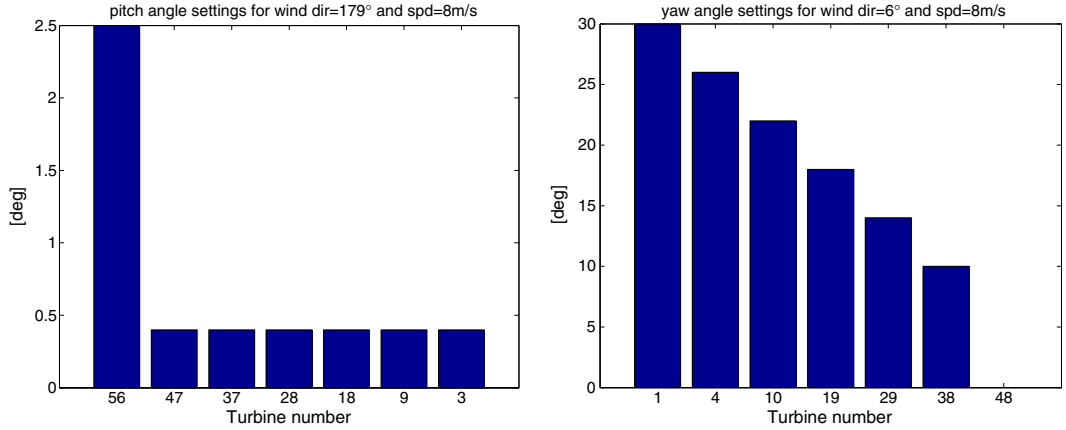


FIGURE 4 Optimized pitch angles (left) and yaw misalignments for the row of wind turbines T1 and T48 for wind speed 8 m/s and wind directions of 179° and 6°, respectively [Colour figure can be viewed at wileyonlinelibrary.com]

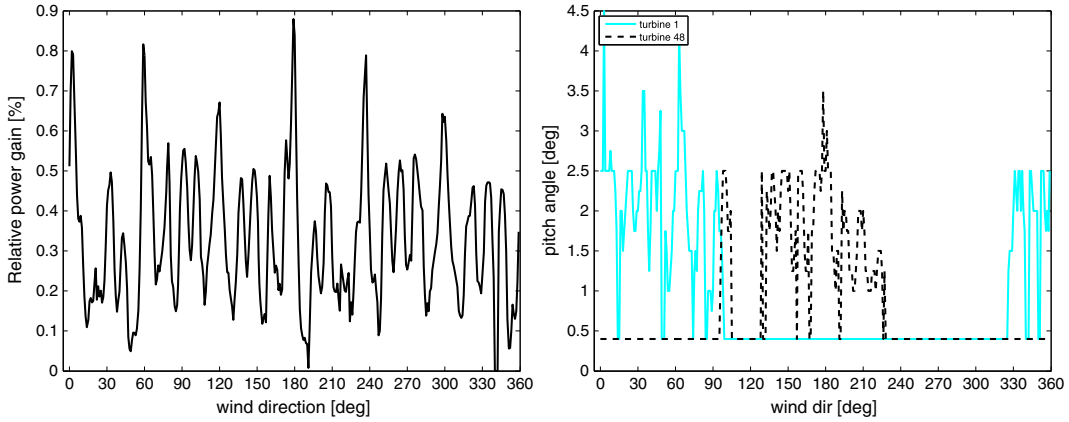


FIGURE 5 Relative power production increase and pitch angle bias for pitch-based active wake control as a function of the wind direction [Colour figure can be viewed at wileyonlinelibrary.com]

T1 and T48. Clearly, in a real-life implementation, the calculated pitch angles will be smoothed out by removing the sharp changes in them. This will only have a small effect on the pitch-based AWC performance.

As mentioned already, the pitch-based AWC strategy cannot be applied to the lifetime optimization problem because the loads database does not include pitch angle variations. For this reason, only yearly power production maximization is performed for pitch-based AWC. Still, it remains interesting to inspect the loads by neglecting the pitch-based AWC pitch basis. For this purpose, Table 2 is provided, which shows the impact of AWC on the lifetime fatigue loads of all turbines in the farm (all wind speeds and directions are considered). The table shows, for the considered components, the reduction of the annual fatigue loading averaged over the turbines (denoted as Load Avg in Table 2) and the corresponding lifetime extension factor (Lifetime Avg), as well as the worst-case (the smallest one over the turbines) load reduction (Load WC) and corresponding lifetime extension factor (Lifetime WC). It can be observed that, even though the loads reductions seem small, the lifetime extension factors are interesting. Indeed, even if the loads on the considered components are assumed equally important, it appears from the table that the turbines in the farm can be operated on the average 0.32% longer than 20 years before the tower reaches its fatigue lifetime. The lifetime energy gain would then be 0.65%.

Looking at the worst-case loading among all turbines, a minor increase is observed for the tower base moment (last 2 columns in the table).

Notice also that the lifetime extension factors for the tower bottom are the lowest, much lower than those for the other considered components. This suggests that there might also be important advantage with respect to corrective maintenance costs, since these loads reductions are expected to reduce the failure rates. However, this benefit is difficult to evaluate numerically and falls outside the scope of this paper.

To get more insight into the loads effects, consider Figure 6 that depicts the tower fore-aft loading as function of the wind direction of 1 single turbine, ie, the one that gets the highest lifetime loading (T27). All wind speeds are included in the figure using the wind speed distribution. It can be seen from the figure that this turbine does not receive a significant load reduction at any wind direction. This is because T27 is located at the center of the wind farm, operating at high turbulence intensity levels both with and without pitch-based AWC. It must be pointed out that there could be other turbines in the farm that do get (slightly) increased loading in some conditions, but their lifetime loads are still lower that that of T27. For instance,

TABLE 2 Loads reduction and lifetime extension per component for the case of single-direction pitch-based active wake control (AWC) for yearly power optimization

Component	Location	Load Avg, %	Lifetime Avg, %	Load WC, %	Lifetime WC, %
Tower fore-aft moment	Bottom	0.079	0.319	−0.035	−0.140
Tower torsion moment	Bottom	0.565	2.284	0.468	1.895
Tower fore-aft moment	Top	0.513	2.070	0.436	1.762
Torque on drive train support		0.376	2.673	0.493	3.524
Rotor-shaft tilt moment		0.241	1.703	0.259	1.832
Rotor-shaft yaw moment		0.256	1.806	0.254	1.795
Axial shaft force at hub		0.285	2.013	0.284	2.011
Blade 1 root flap moment		0.168	1.864	0.152	1.690
Blade 2 root flap moment		0.149	1.659	0.110	1.220
Blade 3 root flap moment		0.131	1.454	0.110	1.221

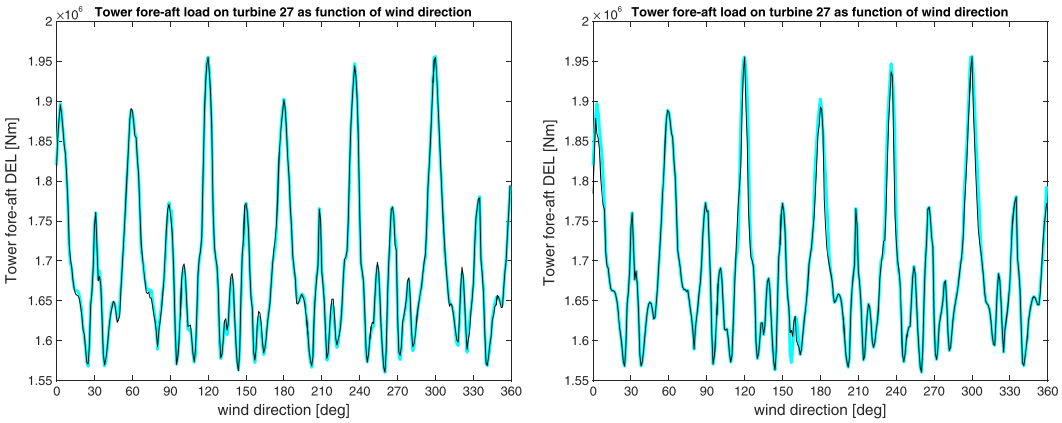


FIGURE 6 Fatigue load on tower fore-aft bottom moment as a function of the wind direction (Weibull distribution included) for the worst-loaded turbine (T27) under pitch-based active wake control (left) and yaw-based active wake control for lifetime power optimization (right) [Colour figure can be viewed at wileyonlinelibrary.com]

notice that the turbulence intensity plot in Figure 3 (right) indicates that the third turbine in the row (T37) receives slightly increased turbulence level at the considered wind conditions which, in combination with the increased wind velocity, suggests increased loading for the considered wind conditions.

4.2.3 | Yaw-based AWC for yearly power optimization

Next, the results from the yaw-based AWC optimization for maximizing the yearly power production (using the single direction approach) are discussed for the Princess Amalia Wind Farm.

Let us again start by considering 1 wind direction (6°) and 1 single row of wind turbines (T1-T48). For this row, Figure 7 gives the power productions and the turbulence intensity, both in the reference case (blue bars) and with yaw-based AWC (red bars). The yaw misalignment settings are given in the right plot of Figure 4. Notice that the chosen wind direction is not exactly parallel to the row orientation, as in such partial wake situations the benefits from wake redirection by yaw-based AWC are largest. As can be seen from Figure 7, the leading turbine produces less power because of the introduced yaw misalignment, while the remaining turbines increase their power production because of the higher wind velocity. As a result, the single-direction yaw-based AWC increases the total power production for row of wind turbines T1 to T48 with 13.3% for the considered wind direction (6°) and wind speed (8 m/s). Looking at the whole park, this benefit reduces to 11.4% more power (still for 1 single wind direction and speed). As AWC operates at below rated farm power, using the wind speed distribution, the yearly increase of the power production for the whole wind farm for this particular wind direction drops to 6.9%. The relative power gain is expressed versus the wind direction in Figure 8 (left), wherein the wind speed distribution is included. Weighing the relative power increase per wind direction with the wind direction probabilities, the total yearly power increase for the whole farm due to single-direction yaw-based AWC becomes around 1.28%. In terms of wake losses, around 11.5% are regained. The plot on the right in Figure 8 shows the yaw misalignment for turbines as function of the wind direction for turbines T23 and 29. Large misalignments of up to 40° are observed.

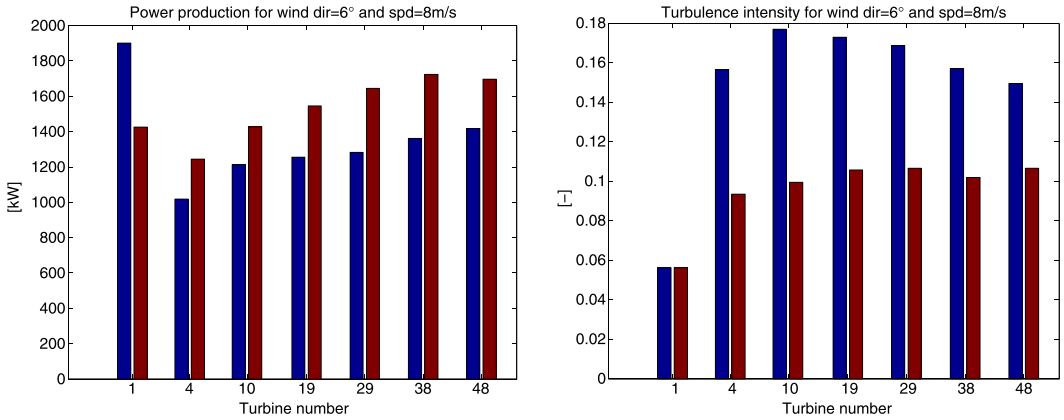


FIGURE 7 Power production (left) and turbulence intensity (right) for the row of wind turbines T1 to T48 for wind direction 6° and wind speed 8 m/s. The blue (left) bars represent the reference case, while the red (right) bars represent yaw-based active wake control results [Colour figure can be viewed at wileyonlinelibrary.com]

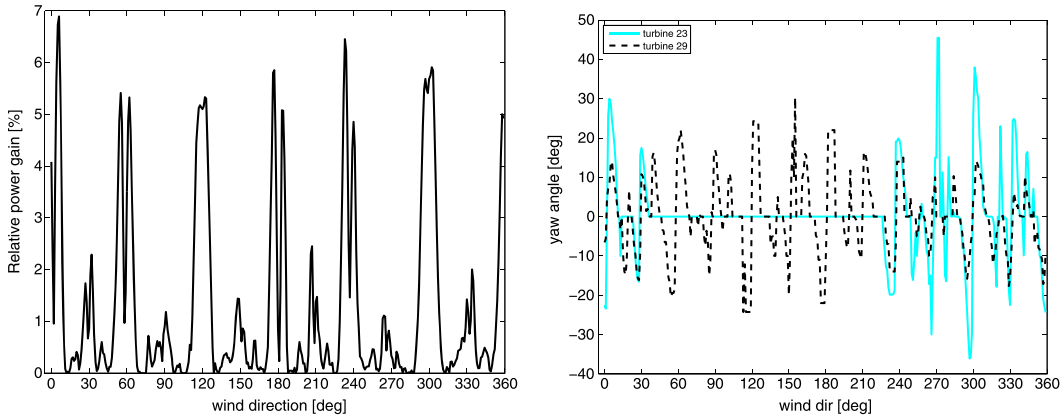


FIGURE 8 Relative power production increase (left) and yaw misalignment (right) for yaw-based active wake control as a function of the wind direction [Colour figure can be viewed at wileyonlinelibrary.com]

4.2.4 | Yaw-based AWC for lifetime power optimization

Finally, the results from the yaw-based AWC optimization for maximizing the lifetime power production (using the sector-based approach) are discussed for the Princess Amalia Wind Farm.

In terms of the yearly power production, the lifetime optimization results in a relative gain of 0.61%, a result less attractive than that achieved when the yearly power is directly maximized (which was 1.28%). However, the added value of the lifetime power optimization becomes clear when the loads are considered. To this end, Table 3 is provided. The table shows, for each of the considered components (see Section (3)), the reduction of the yearly fatigue loading averaged over the turbines (denoted as Load Avg in the table) and the corresponding lifetime extension factor (Lifetime Avg), as well as the worst-case (the smallest one over the turbines) load reduction (Load WC) and corresponding lifetime extension factor (Lifetime WC). It is important to see that the average loading over the turbines reduces for all components considered. Even though the loads reductions seem small in terms of percentages, the corresponding lifetime extension factors are more significant. For instance, the worst-case tower bottom moment fatigue loading is calculated to be just 0.34%. However, the corresponding lifetime extension is 1.38%, which is not negligible. With it, the total lifetime energy gain becomes as much as 2.0%.

Similar to the case with pitch-based AWC optimization, but even more pronounced here, the lifetime extension factors for the tower bottom are the lowest, much lower than those for the other considered components. This suggests that there might also be important advantages with respect to corrective maintenance costs, since the significant loads reductions there imply lower failure rates. Again, a quantitative analysis of this benefit is difficult and out of the scope of this study.

In order to get a better insight into the loads, let's look into the loads on the most loaded wind turbine (T27). The tower fore-aft loading is depicated on Figure 6 (right) as function of the wind direction. The wind speeds are included via the wind speed distribution. It can be seen from the figure that this turbine receives slightly increased loading at some wind directions (for instance, just above 150°). However, the fore-aft loading is most significant for the wind directions aligned with a row of turbines (peaks in the figure), which are the main contributors to the lifetime fatigue loading.

TABLE 3 Loads reduction and lifetime extension per component for the case of yaw-based active wake control (AWC) for lifetime power optimization

Component	Location	Load Avg, %	Lifetime Avg, %	Load WC, %	Lifetime WC, %
Tower fore-aft moment	Bottom	0.068	0.276	0.342	1.380
Tower torsion moment	Bottom	2.074	8.614	3.500	15.318
Tower fore-aft moment	Top	2.053	8.527	3.523	15.425
Torque on drive train support		2.057	15.555	3.087	24.540
Rotor-shaft tilt moment		2.423	18.590	3.605	29.310
Rotor-shaft yaw moment		2.430	18.635	3.605	29.303
Axial shaft force at hub		1.862	13.960	2.762	21.657
Blade 1 root flap moment		1.855	22.819	2.098	26.271
Blade 2 root flap moment		2.048	25.610	2.368	30.164
Blade 3 root flap moment		1.848	22.782	2.169	27.273

TABLE 4 Types of active wake control (AWC) optimizations and loads analysis performed per farm (SINGLE, single-direction based; LA, loads analysis; SECTOR, sector based)

Wind Farm	Pitch-Based AWC		Yaw-Based AWC		
	SINGLE	LA (limited)	SINGLE	LA	SECTOR
Wind farm 1	✓				
Wind farm 2	✓		✓		
Wind farm 3	✓	✓	✓	✓	✓
Princess Amalia	✓	✓	✓	✓	✓
Wind farm 5	✓	✓	✓	✓	
Wind farm 6	✓	✓	✓	✓	

At these directions the load is reduced under AWC, resulting in an overall decrease of the lifetime loading on the tower. Again, even though some other turbine might experience some load increase at some conditions, its lifetime loading will still be lower than that of the worst loaded T27.

4.3 | AWC benefits for the other considered farms

In this section, a summary is presented of the results obtained for the other wind farms, described in Section 4.1, and some interesting observations are made.

As explained in Section 3, the described AWC optimization allows to either maximize the yearly power production, and therefore disregarding the loads (single direction based optimization), or maximize the lifetime power production by including the loads (sector-based optimization). In principle, both approaches are applicable to both pitch-based and yaw-based AWC. However, the sector-based optimization requires the loads module, described in Section 3.1. Since for some of the farms summarized in Table 1 the (single-direction) AWC optimizations have been performed before the loads module was developed, sector-based optimization and loads analysis results are only presented for some of the farms. Moreover, as explained in Section 3.2, partial-load pitch angle variations are not contained in the loads module and, therefore, their effect on the loads is not included in the analysis. For this reason, sector-based AWC is only applied to yaw-based AWC. Nevertheless, for some of the wind farms, loads analysis is also performed for pitch-based AWC anyway, even though it excludes the effect of the pitch angle offset on the loads. Table 4 summarizes the types of AWC optimizations, and loads analysis, performed for the different farms (SINGLE: single direction based optimization for the yearly power production, SECTOR: sector-based optimization for the lifetime power production including loads, LA: loads analysis).

The results for pitch-based and yaw-based AWC per wind farm are summarized in Tables 5 and 6, respectively. Each of these 2 tables reports, per wind farm, the relative power increase, the lifetime extension factor, and the wake loss reduction factor. The power increase is given in 4 different forms (the first 4 columns in the tables):

1. Single Row represents the power increase of 1 single row of turbines in the farm for the most beneficial wind direction and wind speed.
2. Single Direction gives the power production increase of the whole wind farm and all wind speeds (wind speed distribution included) for the most beneficial wind direction.
3. Yearly is the yearly power production increase of the whole farm (ie, effect of loads on the lifetime are not considered).
4. Lifetime is the lifetime power production increase of the whole farm, wherein the effect of the worst-case fatigue load reduction over all components and all wind turbines is considered to determine the lifetime extension factor. More specifically, for each wind turbine, the loads on

TABLE 5 Pitch-based AWC results

Wind farm	Power increase, %				Lifetime extension, %	Wake loss reduction, %
	Single row	Single direction	Yearly	Lifetime		
Wind farm 1	3.3	0.36	0.024	0.024	…	0.25
Wind farm 2	4.23	0.55	0.13	0.13	…	1.25
Wind farm 3	7.8	3.3	0.37	1.25	0.88	3.56
Princess Amalia	2.2	0.88	0.33	0.19	–0.14	2.97
Wind farm 5	6.77	1.31	0.083	0.186	0.10	0.84
Wind farm 6	4.99	1.40	0.62	2.09	1.46	4.88

TABLE 6 Yaw-based AWC results

Wind farm	Power Increase, %				Lifetime extension, %	Wake loss reduction, %
	Single row	Single direction	Yearly	Lifetime		
Wind farm 1	…	…	…	…	…	…
Wind farm 2	16.3	5.3	0.63	0.63	…	6.06
Wind farm 3	4.47	2.28	0.41	1.31	0.90	3.95
Princess Amalia	13.3	6.89	1.28	1.89	0.6	11.5
Wind farm 5	10.1	3.14	0.10	0.14	0.038	1.0
Wind farm 6	26.5	10.06	1.15	1.90	0.74	9.14

the selected components are analyzed with and without AWC (Section 3.1), and the fatigue load reduction is calculated per component. The worst-case (ie, the lowest) load reduction over all components is then chosen as indication for the most conservative load reduction per turbine, and the same is done to select the worst-case load reduction over all turbines. This load reduction factor is then used to calculate the corresponding lifetime extension factor using the relationship between lifetime and stress (Section 3.1).

The last 2 rows give the lifetime extension factor and the yearly power production gain divided by the wake losses (ie, the wake loss reduction factor).

4.3.1 | Pitch-based AWC for yearly power optimization

It can be concluded from Table 5 that the benefit from pitch-based AWC largely depends on the distances between the turbines: For average distances above 7D, the yearly power increase remains limited. This is because at larger distances, the effect of reduction of the axial induction upstream on the wind velocity in the wake remains limited because of the higher wake recovery effects. For distances below 7D, it can be concluded that pitch-based AWC regains roughly 1% to 5% of the wake losses, depending on many factors such as layout, turbulence intensity, aerodynamic properties of the turbines, and wind resource. Interestingly, it can also be concluded from Tables 1 and 5 that the effect of the farm parameters wake loss and power density on the yearly power gain by pitch-based AWC is less pronounced (Wind farm 2 has lower power density than, and comparable wake losses to, Wind farm 1, yet the power production benefit from pitch-based AWC is about 5 times higher).

The lifetime power production is also evaluated for some of the farms. However, as already mentioned, the effect of pitch angle variations on the loads are not considered in this analysis. The lifetime extension factors, reported in Table 5, indicate that pitch-based AWC may enable to prolong the lifetime of the farm by up to 0.9% to 1.5% for the Wind farm 3 and Wind farm 6. For Wind farm 5, the lifetime extension factor is less attractive, and for Princess Amalia, it is even slightly negative. This will be analyzed in more details in Section 4.2.

4.3.2 | Yaw-based AWC for yearly power optimization

Table 6 summarizes the results obtained with yaw-based AWC for the considered farms. It can be concluded the yearly power gains with yaw-based AWC are generally higher that those for pitch-based AWC. In addition, the differences between the results for the different farms are less pronounced here (with the exception of Wind farm 5 for which the gain is lower), ranging from 0.41% power increase for Wind farm 3 to 1.28% for Princess Amalia. While pitch-based AWC increases its benefit when the distance between the turbines reduces, this is not the case with yaw-based AWC where too short distances (below 5D-6D) lead to situations in which there is insufficient room for the wake to move aside from the downstream turbines. This becomes clear by comparing the results for Wind farm 2 and Wind farm 3, where because of the somewhat shorter distances in Wind farm 3, the benefit with yaw-based AWC is lower than for Wind farm 2 (compare the single row results in Table 6), while the pitch-based strategy is more beneficial for Wind farm 3.

With respect to the wake loss reduction, it can be observed from the table that typically 4% to 11.5% of the wake losses are regained by applying yaw-based AWC. An exception here is the Wind farm 5 farm because of its irregular layout that makes it difficult to divert the wake so that it does not hit downstream turbines.

4.3.3 | Yaw-based AWC for lifetime power optimization

When loads are considered, the results are also encouraging: Yaw-based AWC decreases the lifetime fatigue loads on all considered components, leading to a lifetime extension factor of 0.6% to 0.9% in most cases (the exception, Wind farm 5, is not surprising given the very small gain). Combined with the yearly power gain, the lifetime extension results in an increase of the lifetime energy production of 1% to 2%.

While it might seem reasonable to expect the loads to increase because of the introduced yaw misalignment, it turns out that yaw-based AWC actually can reduce the lifetime fatigue loads significantly. Indeed, by itself, yaw misalignment might increase the fatigue loads somewhat, but the wake effects turn out to have a much higher effect on the fatigue loads[41] so that by reducing the wake effects, a lifetime fatigue loads reduction is achievable. This is best explained by considering 1 single row of turbines with the wind direction approximately in line with the row. Yaw-based AWC would in this case misalign all turbines but the last one, which may (or may not [42]) result in loads increase. The wakes, however, will be largely moved aside from the row, resulting in all turbines receiving a large portion of the free (undisturbed by the wake) air flow. The wind velocity will be higher but the main loads driver, the turbulence intensity, is lower. Of course, the leading wind turbine will not feel a difference in the incoming air turbulence so it might end up with a loads increase for some specific wind directions, but when the wind comes from the opposite direction, the loads on that turbine will decrease even more. In the next section, a more in-depth analysis is presented for the Princess Amalia wind farm.

5 | CONCLUSIONS

This paper presented a detailed study on the benefits of wind farm wake mitigation control strategies in terms of power production increase and load reduction. Both pitch-based (axial induction control) and yaw-based AWC (wake redirection) are considered in the analysis. The study makes use of ECN's tool FarmFlow, a very accurate wind farm wake model with reasonable computational complexity. FarmFlow is extended with a loads module that uses FarmFlow output and provide the fatigue loads on a selected set of wind turbine components. An optimization algorithm is developed to optimize the power production of the wind farm. The optimization criterion is selected so as to enable inclusion of the fatigue loads. To avoid modeling the effect of loads on the O&M costs, the effect of the loads on the lifetime of the selected turbine components is considered. Loads reduction is then included into the optimization through its corresponding lifetime extension factor. This formulation allows to easily distinguish between optimization of the yearly power production (disregarding the loads), and lifetime optimization (including power and loads).

Both yearly and (loads-based) lifetime power optimizations are performed for several full-scale commercial wind farms with different scale, layouts and turbine sizes, and the results are compared. The results indicate that pitch-based AWC is primarily interesting for wind farms with turbine distances of up to 6D to 7D, for which 1% to 4% of the wake losses of the wind farm are regained yearly. In addition, it is shown that fatigue loads can also be reduced to an extend that corresponds, in the most conservative (worst case), to a lifetime extension of 1% and even more. With respect to yaw-based AWC, the benefits are even higher. Firstly, yaw-based AWC is shown to have added value in terms of yearly power production increase even for farms with larger distances (>13D analyzed) between the wind turbines. Secondly, in terms of wake loss reduction factor, the results seem to be rather insensitive to the particular farm layout and properties: 4% to 6% of the wake losses are regained on yearly basis by yaw-based AWC. Still, the most important conclusion from the study is that yaw-based AWC not only does not increase the loads on the considered components but actually is capable of reducing those to an extend that suggests lifetime extension of up to 1.5%.

An even higher benefit could be achieved by combining pitch-based and yaw-based AWC wind direction wise, ie, by applying for each wind direction the strategy that gives the highest gain. This combination becomes interesting mostly for farms with regular layouts. In that case, pitch-based AWC will be operated for wind directions well aligned (within a narrow sector) with the rows of turbines, and yaw-based AWC will take over when the wind comes at an angle with respect to the rows. For wind farms with symmetrical (grid-like) layouts, combining pitch-based and yaw-based AWC could achieve almost the sum of the power production benefits of the 2 separate strategies.

Finally, it must be pointed out that even though the used wake modeling software FarmFlow is very accurate in predicting the power productions of individual wind turbines in a wind farm, it cannot perform simulations with time-varying wind speed and wind direction. As a result, the analysis results presented are static, and the practical benefits from AWC might turn out to be lower than predicted. The reason for that is that in real-life, the wind resource varies substantially in time and space, and the AWC settings, and especially the rotor yaw misalignment angles, cannot be modified as fast as the wind changes because of the very slow dynamics of the yaw system. Future research will be aimed at optimizing the AWC algorithm in dynamic environments to get the best balance between AWC performance and pitch and yaw actuator duty.

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